

## 6.11 Projection in the population: best linear prediction

So far the geometry has lived in observation space  $\mathbb{R}^n$ , and every statement was about one fixed data vector. There is a parallel geometry one level up, in which the vectors are *random variables* and the inner product is an expectation. In that geometry, least squares estimates something even when the linear model is false, namely the population projection of  $Y$  onto a span of regressors. In this section  $\mathbf{X}$  is a *random* vector of regressors, not the model matrix  $\mathbf{X}$ . The conditional expectation turns out to be a projection too. This section sets up the correspondence. [Chapter 14](#) develops prediction theory from it, and [Chapter 19](#) uses it to study what least squares estimates under misspecification.

### 6.11.1 Random variables as vectors

Consider random variables with finite second moments on a common probability space. They form a vector space, and

$$\langle U, V \rangle = E(UV), \quad \|U\| = (E U^2)^{1/2}$$

is an inner product, provided we identify variables that are equal with probability one, so that  $\|U\| = 0$  forces  $U = 0$ . This space is usually written  $L^2$ . Two random variables are orthogonal when  $E(UV) = 0$ . For variables with mean zero, this is the same as being uncorrelated. The Cauchy–Schwarz inequality Eq. 6.2.2 becomes the statement that correlations lie in  $[-1, 1]$ . The *cosine* of the angle between two centred random variables is their correlation.

For a random  $k$ -vector  $\mathbf{X}$  with finite second moments, the set  $\mathcal{L} = \{a + \mathbf{b}^\top \mathbf{X} : a \in \mathbb{R}, \mathbf{b} \in \mathbb{R}^k\}$  is a subspace of dimension at most  $k + 1$ . The proofs of Proposition 6.2.2, Theorem 6.2.3 and Theorem 6.3.1 used only the axioms of an inner product and a finite spanning set, so they apply unchanged. Every  $Y$  with  $EY^2 < \infty$  has a unique nearest point in  $\mathcal{L}$ , characterized by orthogonality of the error to  $\mathcal{L}$ .

#### Definition 6.11.1. Best linear predictor

The **best linear predictor** of  $Y$  given  $\mathbf{X}$ , written  $L(Y | \mathbf{X})$ , is the projection of  $Y$  onto  $\mathcal{L}$ . It is the  $a + \mathbf{b}^\top \mathbf{X}$  that minimizes the mean squared error  $E(Y - a - \mathbf{b}^\top \mathbf{X})^2$ .

#### Theorem 6.11.1

Let  $\boldsymbol{\mu}_X = E\mathbf{X}$ ,  $\mu_Y = EY$ ,  $\boldsymbol{\Sigma}_{XX} = \text{Cov}(\mathbf{X})$  and  $\boldsymbol{\sigma}_{XY} = \text{Cov}(\mathbf{X}, Y)$ , and suppose  $\boldsymbol{\Sigma}_{XX}$  is positive definite. Then

$$L(Y | \mathbf{X}) = \alpha^* + \boldsymbol{\beta}^{*\top} \mathbf{X}, \quad \boldsymbol{\beta}^* = \boldsymbol{\Sigma}_{XX}^{-1} \boldsymbol{\sigma}_{XY}, \quad \alpha^* = \mu_Y - \boldsymbol{\beta}^{*\top} \boldsymbol{\mu}_X.$$

The prediction error  $U = Y - L(Y | \mathbf{X})$  has mean zero and is uncorrelated with every component of  $\mathbf{X}$ . Moreover

$$\text{Var}(Y) = \text{Var}(L(Y | \mathbf{X})) + E U^2, \quad \frac{\text{Var}(L(Y | \mathbf{X}))}{\text{Var}(Y)} = \frac{\boldsymbol{\sigma}_{XY}^\top \boldsymbol{\Sigma}_{XX}^{-1} \boldsymbol{\sigma}_{XY}}{\text{Var}(Y)}. \quad (6.11.1)$$

*Proof.* Orthogonality of  $U = Y - a - \mathbf{b}^\top \mathbf{X}$  to the spanning set  $\{1, X_1, \dots, X_k\}$  of  $\mathcal{L}$  gives the *population normal equations*

$$E(U) = 0, \quad E(\mathbf{X}U) = \mathbf{0}.$$

The first gives  $a = \mu_Y - \mathbf{b}^\top \boldsymbol{\mu}_X$ . Substituting into the second and subtracting  $\boldsymbol{\mu}_X E(U) = \mathbf{0}$  gives  $\text{Cov}(\mathbf{X}, Y) - \text{Cov}(\mathbf{X})\mathbf{b} = \mathbf{0}$ , so  $\mathbf{b} = \boldsymbol{\Sigma}_{XX}^{-1} \boldsymbol{\sigma}_{XY}$ . By the projection theorem (Theorem 6.3.1) the solution is the unique minimizer. Since  $U$  has mean zero and is uncorrelated with  $\mathbf{X}$ , it is uncorrelated with  $L(Y | \mathbf{X})$ , so the variances add. Finally,  $\text{Var}(\boldsymbol{\beta}^{*\top} \mathbf{X}) = \boldsymbol{\beta}^{*\top} \boldsymbol{\Sigma}_{XX} \boldsymbol{\beta}^* = \boldsymbol{\sigma}_{XY}^\top \boldsymbol{\Sigma}_{XX}^{-1} \boldsymbol{\sigma}_{XY}$ . ■

The ratio in Eq. 6.11.1 is the population *squared multiple correlation*, the population counterpart of  $R^2$ . By the same cosine argument as Theorem 6.7.2, it is the squared correlation between  $Y$  and  $L(Y | \mathbf{X})$ .

## 6.11.2 Conditional expectation is also a projection

The linear span  $\mathcal{L}$  is one subspace of  $L^2$ . A much larger one is  $\mathcal{G} = \{g(\mathbf{X}) : E g(\mathbf{X})^2 < \infty\}$ , all square-integrable functions of  $\mathbf{X}$ . It is infinite-dimensional, but it is closed, and the projection theorem still holds in it. We accept that fact from measure-theoretic probability and only use its conclusion.

### Proposition 6.11.2

Let  $m(\mathbf{X}) = E(Y | \mathbf{X})$  with  $E Y^2 < \infty$ . Then:

- (a)  $m(\mathbf{X})$  is the projection of  $Y$  onto  $\mathcal{G}$ . It minimizes  $E(Y - g(\mathbf{X}))^2$  over  $g$ , and  $E[(Y - m(\mathbf{X}))g(\mathbf{X})] = 0$  for all  $g(\mathbf{X}) \in \mathcal{G}$ ;
- (b)  $L(Y | \mathbf{X}) = L(m(\mathbf{X}) | \mathbf{X})$ : the best linear predictor of  $Y$  equals the best linear predictor of the regression function;
- (c)  $E(Y - L(Y | \mathbf{X}))^2 = E(Y - m(\mathbf{X}))^2 + E(m(\mathbf{X}) - L(Y | \mathbf{X}))^2$ ;
- (d) if  $m$  is affine,  $m(\mathbf{X}) = L(Y | \mathbf{X})$ .

*Proof.*

- (a) The orthogonality property follows from the tower rule:  $E[(Y - m(\mathbf{X}))g(\mathbf{X})] = E[g(\mathbf{X}) E(Y - m(\mathbf{X}) | \mathbf{X})] = 0$ . Orthogonality to the subspace characterizes the nearest point, by the argument of Theorem 6.3.1, which needs only Pythagoras.
- (b) This is Theorem 6.5.1(a) in  $L^2$ . Since  $\mathcal{L} \subseteq \mathcal{G}$ , projecting onto  $\mathcal{L}$  can be done in two stages.
- (c)  $Y - L = (Y - m) + (m - L)$ , and  $m - L \in \mathcal{G}$  is orthogonal to  $Y - m$  by (a).
- (d) If  $m \in \mathcal{L}$ , the second term in (c) is zero. ■

Part (c) is the population version of Figure 6.5.1. The error of the best linear predictor splits

into two orthogonal parts. One is irreducible noise, which no function of  $\mathbf{X}$  can remove. The other is *approximation error*, the price of insisting on linearity. Chapters in Part IX reduce the second part by enlarging  $\mathcal{L}$  towards  $\mathcal{G}$ .

### 6.11.3 Least squares estimates the population projection

Here is the connection with the rest of the chapter. Given data  $(\mathbf{x}_{(i)}, y_i)$ , put probability  $1/n$  on each row. For this *empirical distribution*, the  $L^2$  inner product of two variables is  $n^{-1} \sum_i u_i v_i = n^{-1} \mathbf{u}^\top \mathbf{v}$ , which is the Euclidean inner product of observation space up to the factor  $1/n$ . So the best linear predictor under the empirical distribution *is* the least squares fit, and the population normal equations become the normal equations Eq. 6.4.1. Observation-space geometry is  $L^2$  geometry for the empirical distribution.

As the sample grows, the empirical distribution approaches the population distribution, and the projections converge with it.

#### Proposition 6.11.3. Consistency for the projection

Let  $(\mathbf{X}_i, Y_i)$ ,  $i = 1, 2, \dots$ , be independent copies of  $(\mathbf{X}, Y)$  with finite second moments and  $\Sigma_{XX}$  positive definite. Let  $(\hat{\alpha}_n, \hat{\beta}_n)$  be the least squares coefficients from the first  $n$  observations, with an intercept. Then  $(\hat{\alpha}_n, \hat{\beta}_n) \rightarrow (\alpha^*, \beta^*)$  with probability one. No assumption about the form of  $E(Y | \mathbf{X})$ , the distribution of the errors, or their variance is needed.

*Proof.* For large  $n$ ,  $\hat{\beta}_n = \mathbf{S}_{XX}^{-1} \mathbf{s}_{XY}$ , where  $\mathbf{S}_{XX}$  and  $\mathbf{s}_{XY}$  are sample covariances with divisor  $n$  (Section 6.6, centring). By the strong law of large numbers the sample means, second moments and cross-moments converge almost surely to their population values. So  $\mathbf{S}_{XX} \rightarrow \Sigma_{XX}$  and  $\mathbf{s}_{XY} \rightarrow \sigma_{XY}$ . Matrix inversion is continuous at the nonsingular  $\Sigma_{XX}$ , so eventually  $\mathbf{S}_{XX}$  is invertible and  $\hat{\beta}_n \rightarrow \beta^*$ . The intercept follows the same way. ■

#### Example. A curved mean

Let  $X \sim \text{Uniform}(0, 2)$  and  $Y = e^X + \varepsilon$  with independent  $\varepsilon \sim N(0, 0.25)$ . The regression function is not linear, but Theorem 6.11.1 still identifies a best line. A short calculation gives  $\text{Cov}(X, Y) = 1$  and  $\text{Var}(X) = 1/3$ , so the population projection has slope 3.000 and intercept 0.195. The listing fits least squares lines to samples of increasing size. The estimates settle on these values, not on any feature of  $e^x$ . Figure 6.11.1 shows the regression function, the population projection, and a least squares fit.

```
import numpy as np
rng = np.random.default_rng(611)

# X ~ Uniform(0, 2), E(Y | X) = exp(X), Var(Y | X) = 0.25.
# Population projection of Y onto span{1, X}: slope Cov(X, Y)/Var(X), intercept E Y
# - slope * E X.
EX, VarX = 1.0, 4.0 / 12.0
EY = (np.exp(2) - 1) / 2
EXY = (np.exp(2) + 1) / 2

# E exp(X)
# E X exp(X)
```

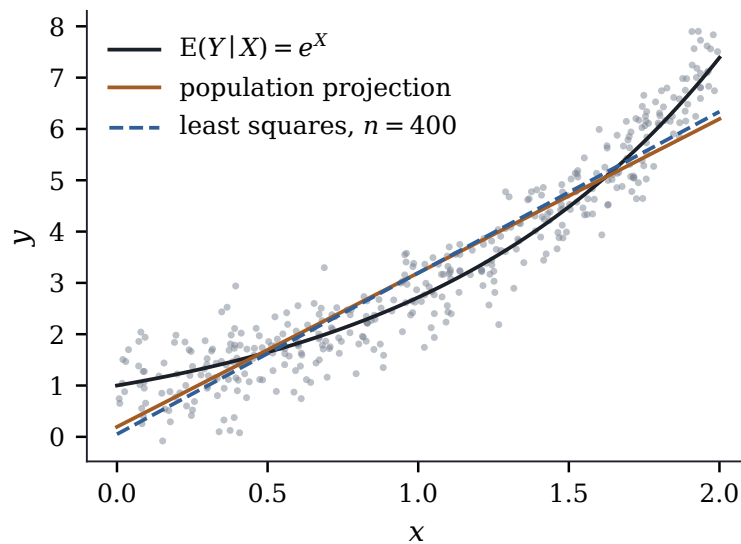


Figure 6.11.1: When the regression function is curved, least squares estimates its best linear approximation in the  $L^2$  sense, the population projection. The approximation error is largest where  $e^x$  bends away from the line.

```
slope_pop = (EXY - EX * EY) / VarX
intercept_pop = EY - slope_pop * EX

for n in [20, 200, 2000, 20000]:
    x = rng.uniform(0, 2, n)
    y = np.exp(x) + rng.normal(0, 0.5, n)
    b, *_ = np.linalg.lstsq(np.column_stack([np.ones(n), x]), y, rcond=None)
    print(f"n = {n:6d}    intercept {b[0]:.3f}    slope {b[1]:.3f}")
print(f"population    intercept {intercept_pop:.3f}    slope {slope_pop:.3f}")
```

[Run this code online](#)

*Remark* (What changes under misspecification). [Proposition 6.11.3](#) is reassuring about the *target*. A least squares line always estimates a well-defined population quantity. It says nothing about the *uncertainty* of the estimate. When  $E(Y | \mathbf{X})$  is not linear, or the variance is not constant, the approximation error  $m(\mathbf{X}) - L(Y | \mathbf{X})$  behaves like extra noise that depends on  $\mathbf{X}$ . The textbook formula  $\sigma^2(\mathbf{X}^\top \mathbf{X})^{-1}$  for the covariance of  $\hat{\beta}$  is then wrong, even in large samples. The “sandwich” covariance estimators of [Chapter 21](#) (Theorem 21.4.2) are designed for exactly this situation.

## 6.11.4 Exercises

### 6.11.4.1 A. Check your understanding

#### Exercise. A1

Verify the numbers in [Example \(A curved mean\)](#) exactly. With  $X \sim \text{Uniform}(0, 2)$  and  $E(Y | X) = e^X$ , evaluate  $EX$ ,  $\text{Var}(X)$ ,  $EY$  and  $E(XY)$  in closed form, and show that the

population projection has slope exactly 3 and intercept exactly  $(e^2 - 7)/2$ . Why does neither depend on the error variance?

*Solution.* The density is  $1/2$  on  $(0, 2)$ , so  $EX = 1$  and  $\text{Var}(X) = 4/12 = 1/3$ . Next  $EY = Ee^X = \frac{1}{2} \int_0^2 e^x dx = (e^2 - 1)/2$ , and since  $\int_0^2 xe^x dx = [(x - 1)e^x]_0^2 = e^2 + 1$ ,  $E(XY) = E(Xe^X) = (e^2 + 1)/2$ . Hence  $\text{Cov}(X, Y) = (e^2 + 1)/2 - (e^2 - 1)/2 = 1$  and  $\beta^* = 1/(1/3) = 3$ , and  $\alpha^* = (e^2 - 1)/2 - 3 = (e^2 - 7)/2$ , the value quoted there. The error enters  $Y$  additively with mean zero and independently of  $X$ , so it changes neither  $\text{Cov}(X, Y)$  nor  $EY - 3EX$ ; it only enlarges  $EU^2$ .

### Exercise. A2

Let  $\mathbf{A}$  be a nonsingular  $k \times k$  matrix and  $\mathbf{d} \in \mathbb{R}^k$ , and let  $a \neq 0$  and  $b$  be scalars. Show from Definition 6.11.1 alone, without the formulas of Theorem 6.11.1, that

$$L(aY + b \mid \mathbf{A}\mathbf{X} + \mathbf{d}) = aL(Y \mid \mathbf{X}) + b.$$

Deduce that the population squared multiple correlation (Proposition 2.5.3) is unchanged by a nonsingular affine change of the regressors.

### Exercise. A3

Let  $X$  take two values only. Show that every function of  $X$  is affine in  $X$ , so that  $\mathcal{G} = \mathcal{L}$  and  $L(Y \mid X) = E(Y \mid X)$ : with a binary regressor the best linear predictor has no approximation error. Extend this to a categorical regressor with  $g$  levels coded by  $g - 1$  indicators, and say which fitted model of Section 6.3 is its sample counterpart.

*Solution.* If  $X \in \{x_0, x_1\}$  with  $x_0 \neq x_1$ , then for any function  $\varphi$  the affine function with  $b = \{\varphi(x_1) - \varphi(x_0)\}/(x_1 - x_0)$  and  $a = \varphi(x_0) - bx_0$  satisfies  $a + bX = \varphi(X)$  at both values, so  $\mathcal{G} \subseteq \mathcal{L}$ ; the reverse inclusion is immediate. Since  $m(X) \in \mathcal{G}$  is then in  $\mathcal{L}$ , Proposition 6.11.2(d) gives  $m = L$ . With  $g$  levels  $l_1, \dots, l_g$ , let  $Z_k = 1\{X = l_k\}$  for  $k = 2, \dots, g$ . Any function of  $X$  is  $\varphi(l_1) + \sum_{k=2}^g \{\varphi(l_k) - \varphi(l_1)\}Z_k$ , affine in the indicators, so again  $\mathcal{G} = \mathcal{L}$  and  $m = L$ . The sample counterpart is the one-way layout of Example (The one-way layout), whose fit replaces each observation by its group mean and so reproduces the conditional mean exactly.

## 6.11.4.2 B. Practice

### Exercise. B1

Split the regressors as  $\mathbf{X} = (X_1, \mathbf{X}_2)$  with  $X_1$  scalar, write  $X_{1.2} = X_1 - L(X_1 \mid \mathbf{X}_2)$  for the part of  $X_1$  not linearly predictable from the rest, and assume  $\text{Var}(X_{1.2}) > 0$ . Show that the coefficient of  $X_1$  in  $L(Y \mid \mathbf{X})$  is

$$\beta_1^* = \frac{\text{Cov}(X_{1.2}, Y)}{\text{Var}(X_{1.2})}.$$

This is Theorem 6.6.2 in the population. What does it say about a regressor that is an exact linear function of the others?

*Solution.* Write  $Y = \alpha^* + \beta_1^* X_1 + \beta_2^{*\top} \mathbf{X}_2 + U$ , where  $U$  has mean zero and is uncorrelated with  $X_1$  and  $\mathbf{X}_2$  (Theorem 6.11.1). The variable  $X_{1.2}$  is an affine function of  $(X_1, \mathbf{X}_2)$ , so it is uncorrelated with  $U$ . By construction it is uncorrelated with every component of  $\mathbf{X}_2$  and with the constant, so  $\text{Cov}(X_{1.2}, \beta_2^{*\top} \mathbf{X}_2) = 0$ . Finally  $\text{Cov}(X_{1.2}, X_1) = \text{Cov}(X_{1.2}, X_{1.2} + L(X_1 | \mathbf{X}_2)) = \text{Var}(X_{1.2})$ , again because the residual of a projection is uncorrelated with the subspace. Collecting terms,  $\text{Cov}(X_{1.2}, Y) = \beta_1^* \text{Var}(X_{1.2})$ . If  $X_1$  is an exact linear function of  $\mathbf{X}_2$ , then  $X_{1.2} = 0$ , the denominator vanishes, and no coefficient is defined: this is the population form of rank deficiency, and  $\Sigma_{XX}$  is then singular.

### Exercise. B2

Show that  $\text{Cov}(\mathbf{X}, Y) = \text{Cov}(\mathbf{X}, m(\mathbf{X}))$  and hence  $\beta^* = \Sigma_{XX}^{-1} \text{Cov}(\mathbf{X}, m(\mathbf{X}))$ : the population projection depends on the conditional distribution of  $Y$  only through the regression function. Which part of Proposition 6.11.2 is this? Use it to say, in one line, why the slope in Example (A curved mean) would be 3 for any error distribution with mean zero, however heteroscedastic.

*Solution.* By the tower rule,  $E[\mathbf{X}\{Y - m(\mathbf{X})\}] = E[\mathbf{X} E\{Y - m(\mathbf{X}) | \mathbf{X}\}] = \mathbf{0}$  and  $E\{Y - m(\mathbf{X})\} = 0$ , so the two covariance vectors agree; now apply Theorem 6.11.1. This is the content of Proposition 6.11.2(b), which says that projecting  $Y$  and projecting  $m(\mathbf{X})$  onto  $\mathcal{L}$  give the same answer. In Example (A curved mean)  $m(x) = e^x$ , so  $\text{Cov}(X, Y) = \text{Cov}(X, e^X) = 1$  and  $\beta^* = 3$ , whatever the noise does, as long as  $E(\varepsilon | X) = 0$  and second moments are finite.

### Exercise. B3

Make the assertion of this section precise: least squares *is* best linear prediction for the empirical distribution. Put probability  $1/n$  on each row  $(\mathbf{x}_{(i)}, y_i)$  of a data set with an intercept, and let  $\mathbf{X}$  and  $Y$  be the resulting random vector and variable.

- Show that  $\Sigma_{XX} = n^{-1} \tilde{\mathbf{X}}_1^\top \tilde{\mathbf{X}}_1$  and  $\sigma_{XY} = n^{-1} \tilde{\mathbf{X}}_1^\top \mathbf{y}$ , where  $\tilde{\mathbf{X}}_1$  is the matrix of centred regressors of Proposition 6.8.2.
- Deduce from Theorem 6.11.1 and Proposition 5.4.3 that  $\beta^*$  and  $\alpha^*$  are the least squares slopes and intercept, and that the ratio in Eq. 6.11.1 is the  $R^2$  of Theorem 6.7.2.
- What corresponds to  $\Sigma_{XX}$  being singular?

*Solution.*

- Under the empirical law,  $E(X_j) = \bar{x}_j$  and  $\text{Cov}(X_j, X_k) = n^{-1} \sum_i (x_{ij} - \bar{x}_j)(x_{ik} - \bar{x}_k)$ , which is the  $(j, k)$  entry of  $n^{-1} \tilde{\mathbf{X}}_1^\top \tilde{\mathbf{X}}_1$ ; likewise  $\text{Cov}(X_j, Y) = n^{-1} \sum_i (x_{ij} - \bar{x}_j)(y_i - \bar{y}) = n^{-1} (\tilde{\mathbf{X}}_1^\top \mathbf{y})_j$ , since the centred column is orthogonal to  $\mathbf{1}$ .
- The factors  $n^{-1}$  cancel, so  $\beta^* = (\tilde{\mathbf{X}}_1^\top \tilde{\mathbf{X}}_1)^{-1} \tilde{\mathbf{X}}_1^\top \mathbf{y}$ , which is the least squares slope vector of Proposition 5.4.3, and  $\alpha^* = \bar{y} - \beta^{*\top} \bar{\mathbf{x}}$  is its intercept. Then  $\text{Var}\{L(Y | \mathbf{X})\} = n^{-1} \|\hat{\mathbf{Y}} - \bar{y}\mathbf{1}\|^2$  and  $\text{Var}(Y) = n^{-1} \|\mathbf{y} - \bar{y}\mathbf{1}\|^2$ , whose ratio is  $R^2$ .

- (c) A singular  $\Sigma_{XX}$  means  $\tilde{\mathbf{X}}_1$  has linearly dependent columns: the design is rank deficient after centring, so the coefficients are not identified, exactly as in Exercise (C2).

### 6.11.4.3 C. Going deeper

#### Exercise. C1

The split of the prediction error in [Proposition 6.11.2\(c\)](#) into noise and approximation error is invisible to first and second moments. Let  $X$  be uniform on  $\{-1, 0, 1\}$  and let  $c > 0$ .

- Let  $Y_1 = X + \varepsilon$  with  $\varepsilon$  independent of  $X$ , mean zero and variance  $c^2 \text{Var}(X^2)$ . Let  $Y_2 = X + c(X^2 - \frac{2}{3})$ , with no noise at all. Show that  $(X, Y_1)$  and  $(X, Y_2)$  have the same means, variances and covariance, hence the same best linear predictor and the same  $EU^2$ .
- Show that in the first pair the approximation error is zero and all of  $EU^2$  is noise, while in the second the noise is zero and all of  $EU^2$  is approximation error. Compute  $EU^2$ .
- What does this say about diagnosing a curved mean from a fitted line and its residual variance alone? Which later chapter takes the question up, and what extra structure does it need?

*Solution.* Throughout,  $EX = 0$ ,  $\text{Var}(X) = 2/3$ ,  $EX^2 = 2/3$ ,  $EX^3 = 0$ ,  $EX^4 = 2/3$ , so  $\text{Var}(X^2) = 2/3 - 4/9 = 2/9$  and  $\text{Cov}(X, X^2) = 0$ .

- Both responses have mean 0:  $EY_1 = 0$  and  $EY_2 = 0 + c(2/3 - 2/3) = 0$ . The covariances agree:  $\text{Cov}(X, Y_1) = \text{Var}(X) = 2/3$  and  $\text{Cov}(X, Y_2) = \text{Var}(X) + c\text{Cov}(X, X^2) = 2/3$ . The variances agree:  $\text{Var}(Y_1) = 2/3 + c^2 \text{Var}(X^2)$  and, since  $X$  and  $X^2 - 2/3$  are uncorrelated,  $\text{Var}(Y_2) = 2/3 + c^2 \text{Var}(X^2)$ . By [Theorem 6.11.1](#) the best linear predictor depends on the joint law only through these moments, so both are  $L(Y | X) = X$ , and  $EU^2 = \text{Var}(Y) - \text{Var}(X) = c^2 \text{Var}(X^2) = 2c^2/9$  in both cases.
- For  $Y_1$ ,  $m_1(X) = X$  is affine, so  $m_1 = L$  and the second term of [Proposition 6.11.2\(c\)](#) is zero: all of  $EU^2$  is  $E(Y - m)^2 = c^2 \text{Var}(X^2)$ . For  $Y_2$ ,  $Y_2$  is a function of  $X$ , so  $m_2 = Y_2$  and  $E(Y - m_2)^2 = 0$ ; the whole of  $EU^2 = E\{m_2(X) - X\}^2 = c^2 E(X^2 - 2/3)^2 = c^2 \text{Var}(X^2)$  is approximation error.
- Nothing in the fitted line, the residual variance or the population  $R^2$  distinguishes pure noise around a straight line from an exactly curved mean; the two are the same to second order. Separating them needs replication or an assumed shape for the alternative, which is what the lack-of-fit tests of [Chapter 14](#) supply: repeated observations at the same value of  $\mathbf{X}$  estimate the noise on its own, and the rest of the residual sum of squares is the curvature.

#### Exercise. C2

A population projection can reverse the sign of a partial effect. Let  $X_2$  be uniform on  $\{-1, 0, 1\}$ , let  $Z \sim N(0, 1)$  be independent of  $X_2$ , and put

$$X_1 = X_2^2 + Z, \quad Y = m(X_1, X_2) = X_1 - 11X_2^2.$$

- Show that  $\Sigma_{XX}$  is diagonal with entries  $11/9$  and  $2/3$ , and that  $\sigma_{XY} = (-11/9, 0)^\top$ .

Deduce  $L(Y | \mathbf{X}) = -6 - X_1$ .

- (b) So the population projection says that  $Y$  falls by one unit for each unit of  $X_1$ , while  $\partial m / \partial x_1 = 1$  at every point: holding  $x_2$  fixed,  $Y$  rises with  $x_1$ . Explain which feature of the design produces the reversal, and check the answer against [Exercise \(B1\)](#).
- (c) Add  $X_2^2$  to the regressor list. Show that the projection then reproduces  $m$  exactly, with coefficient 1 on  $X_1$ .

*Solution.* Write  $W = X_2^2$ , so  $E W = 2/3$ ,  $\text{Var}(W) = 2/9$  and  $\text{Cov}(W, X_2) = E X_2^3 - E W E X_2 = 0$ .

- (a)  $\text{Var}(X_1) = \text{Var}(W) + \text{Var}(Z) = 2/9 + 1 = 11/9$ ,  $\text{Var}(X_2) = 2/3$  and  $\text{Cov}(X_1, X_2) = \text{Cov}(W, X_2) = 0$ , so  $\Sigma_{XX} = \text{diag}(11/9, 2/3)$ , positive definite even though  $X_1$  and  $X_2$  are functionally dependent through  $W$ . With  $Y = W + Z - 11W = Z - 10W$ ,

$$\text{Cov}(X_1, Y) = \text{Cov}(W+Z, Z-10W) = 1 - 10 \text{Var}(W) = 1 - \frac{20}{9} = -\frac{11}{9}, \quad \text{Cov}(X_2, Y) = -10 \text{Cov}(X_2, W)$$

Hence  $\beta^* = (-1, 0)^\top$ , and  $\alpha^* = E Y - \beta^{*\top} \mu_X = -\frac{20}{3} + \frac{2}{3} = -6$ .

- (b) The regressors are uncorrelated, so  $X_{1.2} = X_1 - E X_1$  and [Exercise \(B1\)](#) gives  $\beta_1^* = \text{Cov}(X_1, Y) / \text{Var}(X_1)$  directly. The trouble is that the model space contains  $X_2$  but not  $X_2^2$ , and  $X_2^2$  is uncorrelated with both regressors while being a large part of both  $X_1$  and  $Y$ . Its effect therefore cannot be attributed to  $X_2$ , and lands on  $X_1$ , whose fluctuation it shares. Linear adjustment adjusts only for what is linearly predictable.
- (c) With regressors  $(X_1, X_2, W)$  the function  $m$  is a linear combination of them, so by [Proposition 6.11.2](#)(d) the projection is  $m$  itself:  $L(Y | X_1, X_2, W) = X_1 - 11W$ , with coefficient 1 on  $X_1$  and prediction error zero. (The three regressors have a nonsingular covariance matrix, since  $Z$  makes  $X_1$  vary freely given  $X_2$ .)